# Tele health monitoring for disease management

# (For Coma Patients)

**Project Guide:**
**Dr. P.Grace Kanmani Prince**

**Team Members:**
**Balaji M 41240006**
**Charan K V 41240009**
**Ranjith B 41240028**

# INTRODUCTION

- Coma is a profound unconscious state. The patient in a coma has no physical control over his or her complete body and is ignorant of the world around them, even though their heart is still pumping.
- To preserve the patients' lives in such situations, careful consideration and ongoing observation are needed.
- Having someone supervise a severely ill person these days costs a lot of money and requires a large staff.
- Furthermore, the constant oversight provided by a paramedical assistant can be problematic because human error can happen.
- Vital signs must be taken in critically ill patients and monitoring of patients in a coma differs from that of regular patients.

# Abstract


- A band type multi-sensor health monitoring system using Ardino is developed.
- The three health parameters implemented are heartbeat, movement detection and spo2 .
- Using these three parameters, the patient's condition is being monitored continuously .
- There is no need for a lot of clinical staff to be physically present to check the condition of the patient because there is GSM module to send SMS message.
- Includes an alert system for notifying users when vital signs exceed thresholds and there is a movement.

# OBJECTIVE

- To measure heart rate,spo2, and movement
- Displaying the live data on display.
- Design an appropriate circuit for GSM patient health monitoring system.
- Test, implement and establish communication between the microcontroller, sensors, and GSM.
- Develop a system that is able to send SMS alert messages in case of emergency.

# LITERATURE SURVEY

| S.No | Title/Author/year of publication | Methodology | Results |
|---|---|---|---|
| 1 | **Coma Patient Health Monitoring System**<br>Authors - Subha R, Haritha M et.al.,<br>2020 | In this model, The Plethysmograph is used which records changes in blood volume as the blood vessel beat extends and gets the microvasculature. This sensor is coordinated with the microcontroller. | Realtime monitoring of BP and pulse |
| 3 | **Telemonitoring of Patients With Chronic Traumatic Brain Injury: A Pilot Study**<br>Aldo Mauro et.al.,<br>2021 | a mono-center randomized controlled study, evaluating the clinical evolution of severe TBI patients after ~4 years intervals from admission in two different long-term care facilities using telehealth monitoring | confirm the potential of telehealth for the chronic management of TBI patients and its use |
| 3 | **Analysis and monitoring of coma patients using wearable motion detector system**<br>Authors - SC Koganti et.al., 2015 | Wearable Motion sensor system is used to monitor the body movements such as eye blink movement and hand movement to detect the conscious state of an individual. | Monitoring and alerting the medical person, when the comatose gains consciousness using motion detection system |

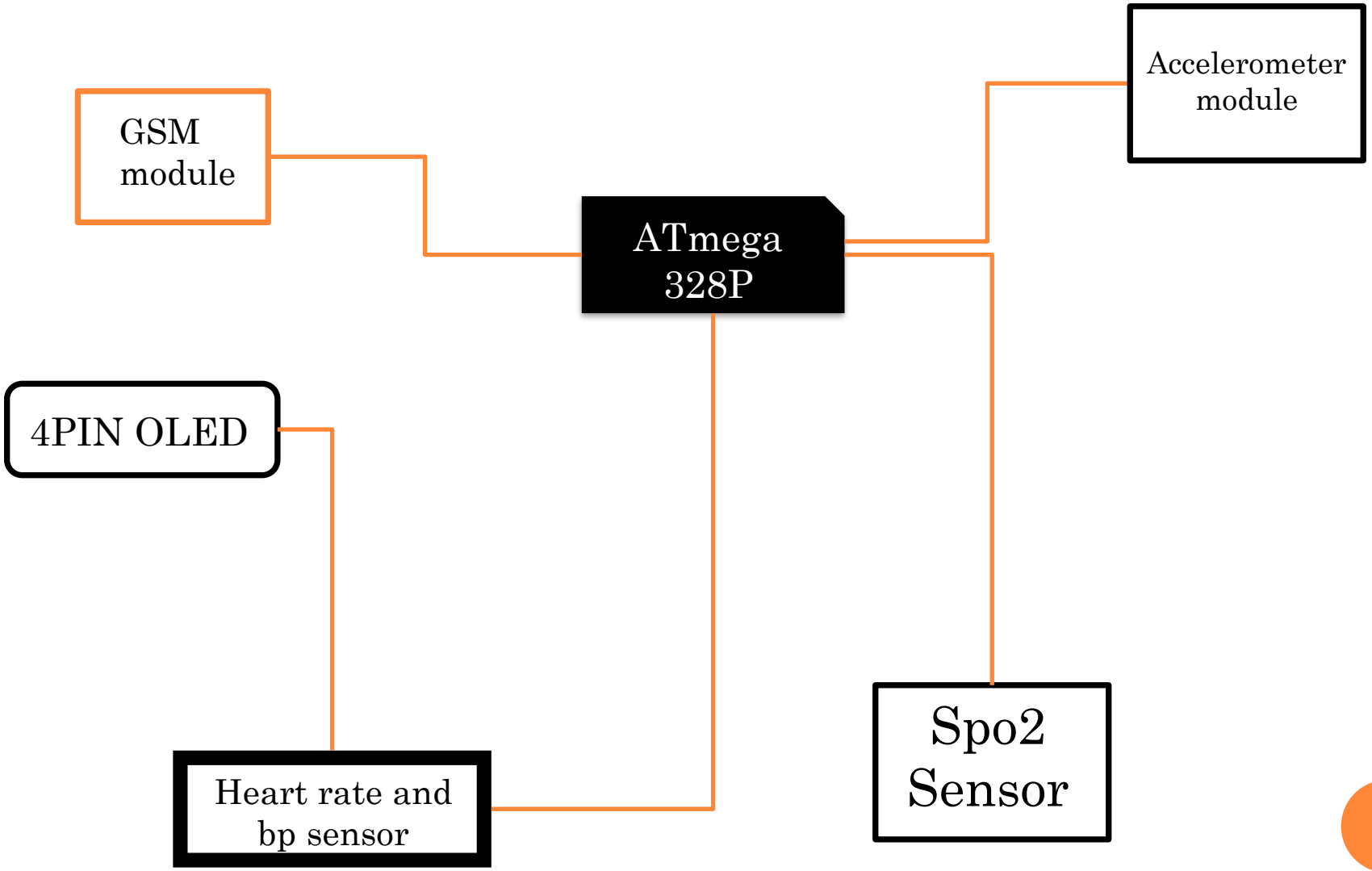
| S.No | Title/Author/year of publication | Methodology | Results |
|---|---|---|---|
| 4 | **Zigbee and GSM based patient health monitoring system** Purnima Puneet Singh et.al., 2014 | In the proposed system, patient's parameters such as ECG, Temperature and Heart Beat will be continuously transmitted and monitored through wireless technology Zigbee | This system do not require the patient to be confined to his bed and allows him to move freely in his room within a specific distance from the doctor's monitor |
| 5 | **Wearable Fall Detection Systems: A Review of Recent Advances.** AUTHORS - Brown,S et.al 2019 | Examined three main categories of fall detection systems—environmental sensing-based, vision-based, and wearable sensor-based—highlighting their features, advantages, and limitations. | Provided an overview of different types of fall detection systems. Discussed the relevance of machine learning in improving fall detection accuracy and minimizing false alarms. |
| 6 | **Machine Learning for Healthcare Wearable Devices: The Big Picture** AUTHOR - Farida Sabry et,al., 2022 | Uses machine learning techniques to learn from human body signals, recorded by wearable devices | potential for early detection and monitoring of health issues using wearable devices. |

| S.No | Title/Author/year of publication | Methodology | Results |
|---|---|---|---|
| 7 | **Comatose Patient Monitoring System Based on (IoT)**<br>Authors - Hussam Ahmed Rammo<br>2023 | The work is divided based on a multi-sensor system, which is applied scientifically and practically on the patient's body including a heart rate sensor, oxygen level in the blood sensor, ECG sensor, and a digital camera for monitoring the patient's eye.. | tangible results in readings of IoT sensors such as ECG sensor with 97.6% accuracy and pulse oximeter sensor with rates up to 96% SPO2 accuracy as well as 83% BMP accuracy, as well as accurate results for digital camera sensor with high accuracy of 94.9% are obtained |
| 8 | **Systematic review of smart health monitoring using deep learning and Artificial intelligence**<br>Author - A.V.L.N. Sujith<br>2021 | integration of blockchain framework improved data security and data privacy of confidential data of patient to prevent the data misuse against patients. Involvement of Deep Learning and Machine learning to analyze health data to achieve multiple targets has helped attain preventive healthcare | This has helped in the early detection of chronic diseases which was not possible recently. To make the services more cost effective and real time, the integration of cloud computing and cloud storage has been implemented. |

# Advantage of our Project over Previous Papers:

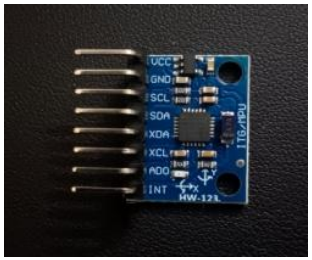
- While the existing systems provide valuable insights into the benefits and challenges of Telehealth this system offers a practical implementation of these concepts.
- By integrating multiple sensors and incorporating alert systems and SMS notifications, this system provides a comprehensive solution for remote patient monitoring.
- It is a compact wearable device which makes its use easier and comfortable for the user .

# BLOCK DIAGRAM



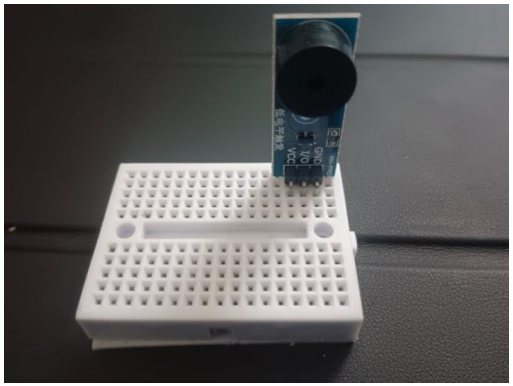
# COMPONENTS

MAX30100
SENSOR

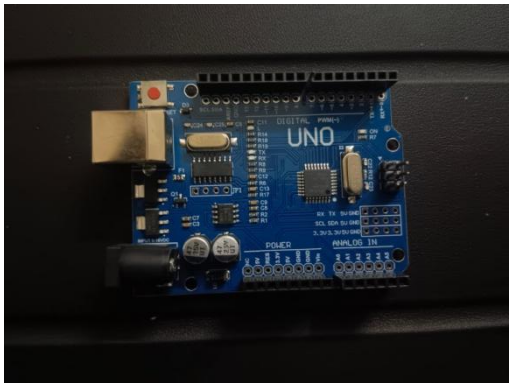



MPU6050
SENSOR

SIM800L
GSM MODULE



BREADBOARD AND
ALARM BUZZER




ARDINO UNO
ATMEGA 328P

# COMPONENTS:

- Arduino board – ATMEGA328P
- Heart rate sensor - MAX30100
- Motion detection sensor - MPU6050 accelerometer
- OLED display
- GSM module SIM800L for sending SMS
- Jumper wires
- Power source - USB cable
- Breadboard
- buzzer

# WORKING

**Max30100 Integration:**

- Measures heart rate and SpO2 and data sent to Arduino for processing.

**MPU6050 Integration:**

- Detects patient movement in 3 axis and sends data to Arduino for analysis.

**Communication Module:**

SIM800L handles GSM communication and sends alerts to designated recipients via SMS.

**Alert System:**

- Buzzer alerts for abnormal heart rhythms or movement.
- SIM800L triggers SMS alerts in emergencies.

**Display Functionality:**

- 4-pin OLED display visualizes real-time data and alerts and is connected to Arduino for data presentation.

**Data Processing:**

- Arduino processes incoming data from sensors and it determines appropriate responses based on predefined thresholds.

**Real-time Monitoring:**

- Continuous monitoring of vital signs and patient movement and Immediate alerts enable timely intervention.

**Feedback Mechanism:**

- Display provides visual feedback to healthcare providers also Audible alerts ensure attention even without visual monitoring

The code initializes the serial communication and the PulseOximeter instance (pox)

# ALGORITHM

Heart Rate Measurement:

- Uses light-based methods (photoplethysmography or PPG).
- Sensors collect data on the changes in blood volume or blood flow.
- PPG sensors emit light into the skin and measure the amount of light absorbed or reflected, which changes with each heartbeat.
- This data is converted into electrical signals.
- Electrical signals are processed to detect individual heartbeats.
- The time between successive peaks is used to calculate the heart rate (beats per minute).

Output

- Final output is the heart rate value, indicating the number of heartbeats per minute.

Accuracy:

- Sensor accuracy can be affected by factors like motion artifacts, ambient light, and sensor placement.

Movement Detection:

- Sensor generates voltage proportional to acceleration.
- Measures acceleration along each axis at a set rate.
- Voltage is converted to digital using ADC.
- Converts digital values to meaningful acceleration units.

Noise Filtering:

- Filters data to remove noise and artifacts.

Output:

- Provides digital acceleration values for further processing.

- SPO2 MEASUREMENT:

- Two light wavelengths (usually red and infrared) are used to determine the amount of oxygen in the blood.
- Oxygenated hemoglobin (HbO2) and deoxygenated hemoglobin (Hb) absorb light differently.
- The ratio of absorbed light at different wavelengths is used to calculate SpO2.
- Electrical signals are processed
- Final output is the SpO2 value, indicating the percentage of oxygen-saturated hemoglobin in the blood.

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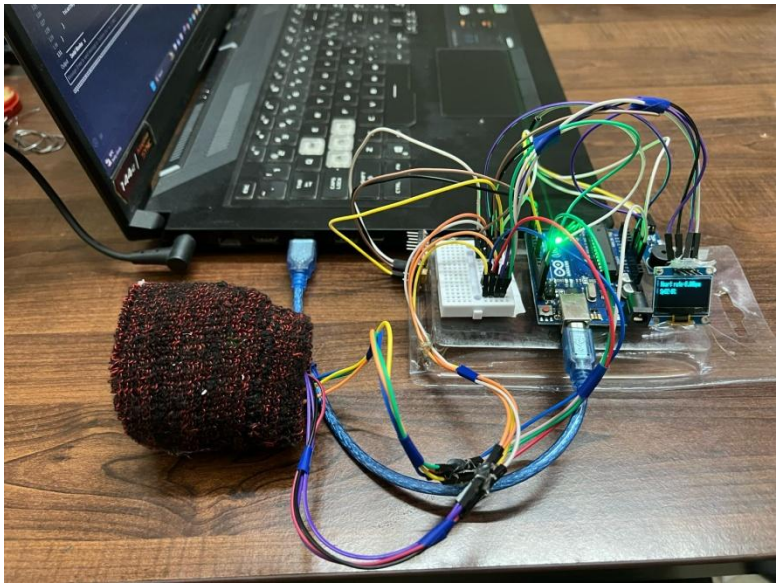
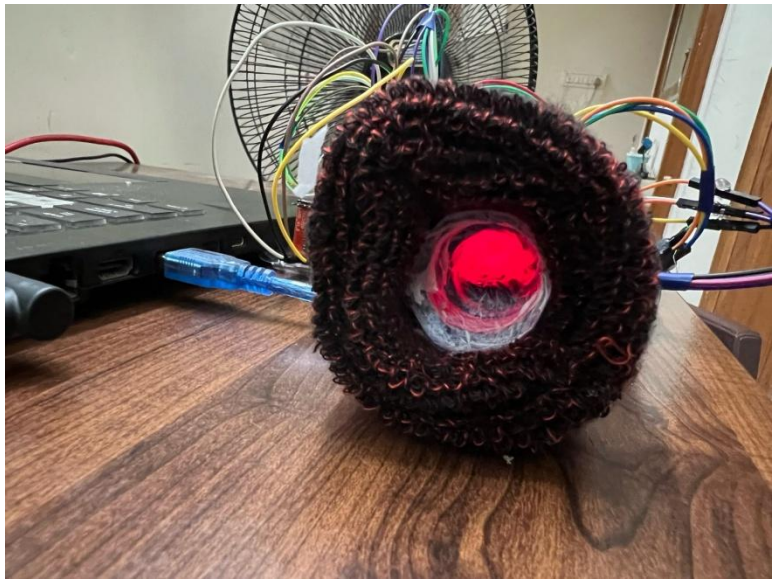
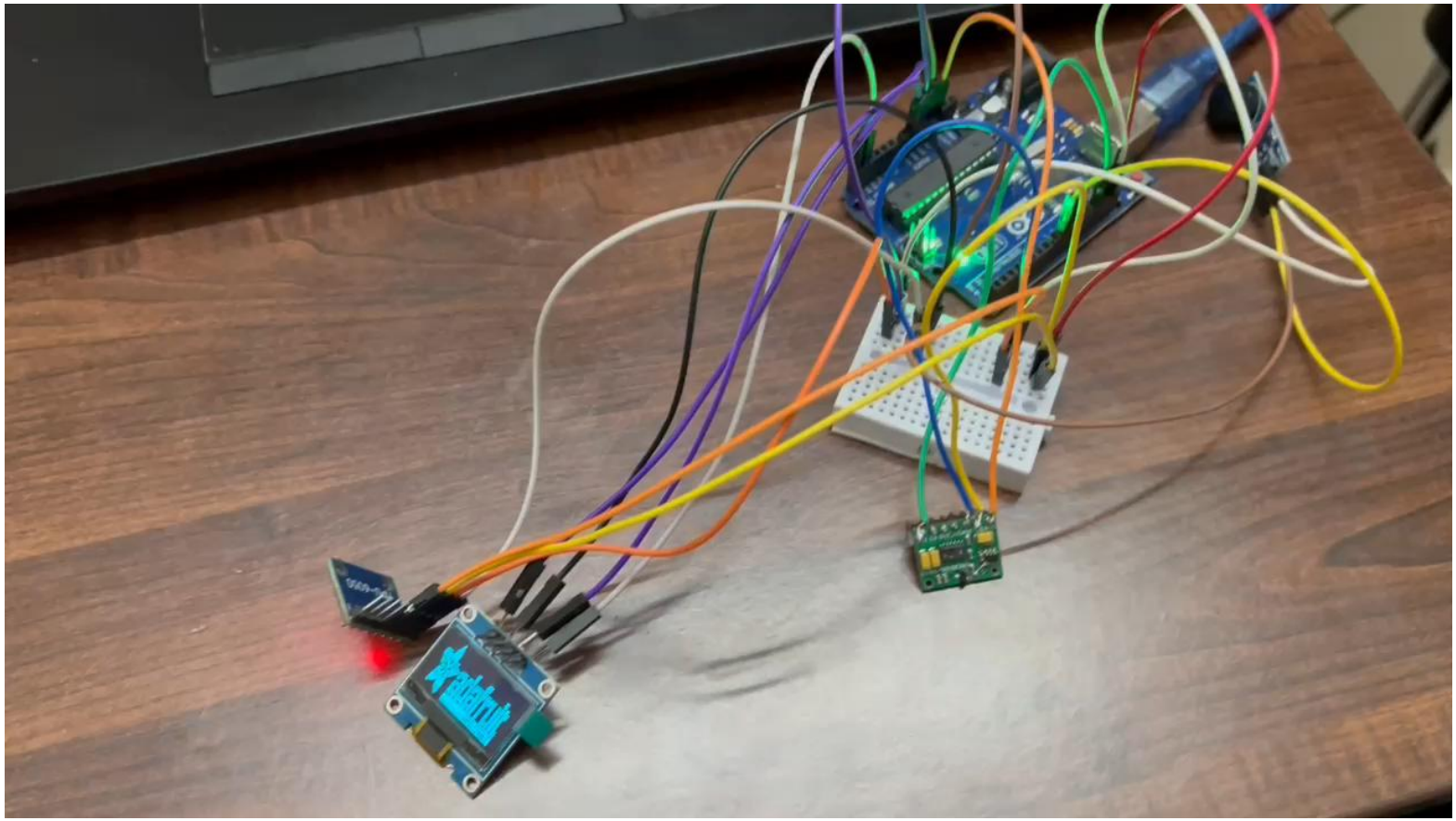
# PROTOTYPE





# PROTOTYPE VIDEO



# Conclusion

- Band based structure make it easy to wear.
- Enables real time data monitoring
- Facilitates early detection of complications.
- Valuable tool for patient care.

THANK YOU